

WEEK3 PRACTICAL

Task 1: Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
rishill10@ASUS16X: ~/practica × + v
rishill10@ASUS16X:~$ mkdir practical3
rishill10@ASUS16X:~$ cd practical3
rishill10@ASUS16X:~/practical3$ nano process.c
rishill10@ASUS16X:~/practical3$ gcc process.c -o process
rishill10@ASUS16X:~/practical3$ ./process

Parent Process
PID = 663
PPID = 337

Child Process
PID = 664
PPID = 663
rishill10@ASUS16X:~/practical3$ |
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Process creation failed.\n");
    }

    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());

        sleep(5);
    }

    else
    {
        printf("\nParent Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());

        wait(NULL);
    }

    return 0;
}
```

Task 2: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
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Parent Process
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rishill10@ASUS16X:~/practical3$ |
```

```
rishill10@ASUS16X: ~/practical x Ubuntu x + v
rishill10@ASUS16X:~$ ps -el
F S UID PID PPID C PRI NI ADDR SZ WCHAN TTY TIME CMD
4 S 0 1 0 0 80 0 - 6025 - ? 00:00:00 systemd
5 S 0 2 1 0 80 0 - 795 - hvc0 00:00:00 init-systemd(Ub
0 S 0 7 2 0 80 0 - 795 - hvc0 00:00:00 init
4 S 0 47 1 0 79 -1 - 12599 - ? 00:00:00 systemd-journal
4 S 989 79 1 0 80 0 - 5635 - ? 00:00:00 systemd-resolve
4 S 0 85 1 0 80 0 - 8855 - ? 00:00:00 systemd-udev
4 S 0 108 1 0 80 0 - 722 - ? 00:00:00 chronyd-starter
4 S 0 109 1 0 80 0 - 1118 - ? 00:00:00 cron
4 S 995 110 1 0 80 0 - 2205 - ? 00:00:00 dbus-daemon
4 S 0 140 1 0 80 0 - 11023 - ? 00:00:00 networkd-dispat
4 S 0 169 1 0 80 0 - 4609 - ? 00:00:00 systemd-logind
4 S 0 175 1 0 80 0 - 448075 - ? 00:00:00 wsl-pro-service
4 S 100 197 1 0 80 0 - 55137 - ? 00:00:00 rsyslogd
4 S 988 245 108 0 80 0 - 5105 - ? 00:00:00 chronyd
1 S 988 246 245 0 80 0 - 3023 - ? 00:00:00 chronyd
4 S 0 280 1 0 80 0 - 30864 - ? 00:00:00 unattended-upgr
4 S 0 283 1 0 80 0 - 1373 - tty1 00:00:00 agetty
5 S 0 334 2 0 80 0 - 796 - ? 00:00:00 SessionLeader
5 S 0 335 334 0 80 0 - 800 - ? 00:00:00 Relay(337)
4 S 1000 337 335 0 80 0 - 1567 do_wai pts/0 00:00:00 bash
4 S 0 338 2 0 80 0 - 2161 - ? 00:00:00 login
4 S 1000 384 1 0 80 0 - 5584 - ? 00:00:00 systemd
5 S 1000 386 384 0 80 0 - 5728 - ? 00:00:00 (sd-pam)
4 S 1000 415 338 0 80 0 - 1534 poll_s pts/1 00:00:00 bash
0 S 1000 764 337 0 80 0 - 692 do_wai pts/0 00:00:00 process
1 S 1000 765 764 0 80 0 - 692 hrtime pts/0 00:00:00 process
5 S 0 766 2 0 80 0 - 796 - ? 00:00:00 SessionLeader
5 S 0 767 766 0 80 0 - 800 - ? 00:00:00 Relay(774)
5 S 0 768 85 0 80 0 - 8856 - ? 00:00:00 (udev-worker)
5 S 0 769 85 0 80 0 - 8856 - ? 00:00:00 (udev-worker)
5 S 0 770 85 0 80 0 - 8856 - ? 00:00:00 (udev-worker)
5 S 0 771 85 0 80 0 - 8856 - ? 00:00:00 (udev-worker)
5 S 0 772 85 0 80 0 - 8856 - ? 00:00:00 (udev-worker)
5 S 0 773 85 0 80 0 - 8856 - ? 00:00:00 (udev-worker)
4 S 1000 774 767 2 80 0 - 1567 do_wai pts/2 00:00:00 bash
0 R 1000 810 774 0 80 0 - 1798 - pts/2 00:00:00 ps
rishill10@ASUS16X:~$
```